2025-10-03 - Valentini (2)

Previous lecture: 2025-10-02 - Valentini (1)
Next lecture: 2025-10-09 - Valentini (3)

$$\frac{\partial n_\alpha}{\partial t} + \nabla \cdot \left(n_\alpha \bar{U}_\alpha\right) = 0$$

$$m_\alpha n_\alpha \left[\frac{\partial \bar{U}}{\partial t} + \bar{U}_\alpha \cdot \nabla \bar{U}_\alpha\right] = -\nabla p_\alpha - q_\alpha n_\alpha \nabla \Phi$$

$$-\nabla^2 \Phi = 4\pi \sum_\alpha q_\alpha n_\alpha$$

$$p_\alpha = n_\alpha k_B T_\alpha$$

$$T_{\alpha 0} = const$$

we linearize the system and we study perturbation around equilibrium

$$n_{i0} = n_{e0} = n_0, \quad \bar{U}_{\alpha 0} = 0, \quad \Phi_0 = 0$$

at equilibrium

$$\implies -\nabla^2 \Phi = 4\pi \sum_\alpha q_\alpha n_\alpha = 4\pi e \left[n_i - n_e\right] = 0 \implies \bar{E}_0 = 0$$

(there are other choices, but we don't lose generalization)

High frequency regime

ions stay fixed, n_{i0}
only equation for electrons

$$\frac{\partial n_e}{\partial t} + \nabla \cdot \left(n_e \bar{U}_e\right) = 0$$

$$m_e n_e \left[\frac{\partial \bar{U}_e}{\partial t} + \bar{U}_e \cdot \nabla \bar{U}_e\right] = -\nabla p_e - q_e n_e \nabla \Phi$$

$$-\nabla^2 \Phi = 4\pi \sum_\alpha q_\alpha n_\alpha$$

$$n_e = n_{e0} + n_1, \quad \bar{U}_e = \bar{U}_{e1}, \quad \Phi = \Phi_1$$

and

$$\frac{\partial}{\partial t}(n_{e0} + n_{e1}) + \nabla \cdot \left((n_{e0} + n_{e1}) \cdot \bar{U}_{e1}\right) = 0$$

$$\frac{\partial}{\partial t}(n_{e0}) + n_{e0} \nabla \cdot \left(\bar{U}_{e1}\right) = 0$$

$$m_e \left(n_{e0} + \cancel{n_{e1}}\right) \left[\frac{\partial \bar{U}_{e1}}{\partial t}\right] = -k_B T_e \nabla \left(\cancel{n_{e0}} + n_{e1}\right) - e \left(n_{e0} + \cancel{n_{e1}}\right) \nabla \Phi_1$$

$$-\nabla^2 \Phi = 4\pi \sum_\alpha q_\alpha n_\alpha$$

we get

$$\begin{cases} \dfrac{\partial}{\partial t}(n_{e1}) + n_{e0}\nabla \cdot \bar{U}_{e1} = 0 & (A) \\ m_e n_{e0}\dfrac{\partial \bar{U}_{e1}}{\partial t} = -k_B T_{e0}\nabla n_{e1} + e n_{e0}\nabla\Phi_1 & (B) \\ \nabla^2\Phi_1 = 4\pi e n_{e1} & (C) \end{cases}$$

we use fourier

$$\begin{cases} n_{e1} = \hat{n}_{e1}\exp\left[i\left(\bar{k}\cdot\bar{r} - \omega t\right)\right] \\ \bar{U}_{e1} = \hat{\bar{U}}_{e1}\exp\left[i\left(\bar{k}\cdot\bar{r} - \omega t\right)\right] \\ \Phi_{e1} = \hat{\Phi}_{e1}\exp\left[i\left(\bar{k}\cdot\bar{r} - \omega t\right)\right] \end{cases}$$

for witch

$$\frac{\partial}{\partial t} \to -i\omega, \qquad \frac{\partial}{\partial r} \to i\bar{k}$$

and

$$\begin{cases} -i\omega\hat{n}_{e1} + n_{e0}\, i\bar{k}\cdot\hat{\bar{U}}_{e1} = 0 & (A) \\ m_e n_{e0}\,(-i\omega)\hat{\bar{U}}_{e1} = -k_B T_{e0} i\bar{k}\, n_{e1} + e n_{e0} i\bar{k}\,\hat{\Phi}_1 & (B) \\ -k^2\hat{\Phi}_1 = 4\pi e\hat{n}_{e1} & (C) \end{cases}$$

Dividing (A) for in_{e0} we have

$$\bar{k}\cdot\hat{\bar{U}}_{e1} = \omega\frac{\hat{n}_{e1}}{n_{e0}}$$

Multiplying (B) for k/i we have

$$-m_e n_{e0}\omega\bar{k}\cdot\hat{\bar{U}}_{e1} = -k_B T_{e0} k^2\hat{n}_{e1} + e n_{e0} k^2\hat{\Phi}_1$$

and using

$$v_{th\,e}^2 = \frac{k_B T_{e0}}{m_e}$$

we have

$$-n_{e0}\omega\bar{k}\cdot\hat{\bar{U}}_{e1} = -v_{th\,e}^2 k^2\hat{n}_{e1} + \frac{e n_{e0}}{m_e}k^2\hat{\Phi}_1$$

now we use (A)

$$-n_{e0}\omega^2\frac{\hat{n}_{e1}}{n_{e0}} = -v_{th\,e}^2 k^2\hat{n}_{e1} + \frac{e n_{e0}}{m_e}k^2\hat{\Phi}_1$$

$$\Longrightarrow \hat{n}_{e1}\left(-\omega^2+k^2 v_{th\,e}^2\right)=\frac{e n_{e0}}{m_e} k^2 \hat{\Phi}_1$$

$$\Longrightarrow \hat{n}_{e1}=\frac{e n_{e0}}{m_e\left(-\omega^2+k^2 v_{th\,e}^2\right)} k^2 \hat{\Phi}_1$$

We use this result in (C)

$$-k^2 \hat{\Phi}_1=\frac{4\pi e^2}{m_e} \frac{n_{e0} k^2 \hat{\Phi}_1}{-\omega^2+k^2 v_{th\,e}^2}$$

using

$$\omega_p^2=\frac{4\pi e^2 n_{e0}}{m_e}$$

we have

$$k^2 \hat{\Phi}_1\left[1-\frac{\omega_p^2}{\omega^2-k^2 v_{th\,e}^2}\right]=0$$

We solve for k
$k=0$ is not useful
we want $\omega^2-k^2 v_{th\,e}^2=\omega_p^2$ and

$$\omega=\pm\sqrt{\omega_p^2+k^2 v_{th\,e}^2}=\pm\omega_p\sqrt{1+\frac{k^2 v_{th\,e}^2}{\omega_p^2}}$$

which is the dispersion relation
also

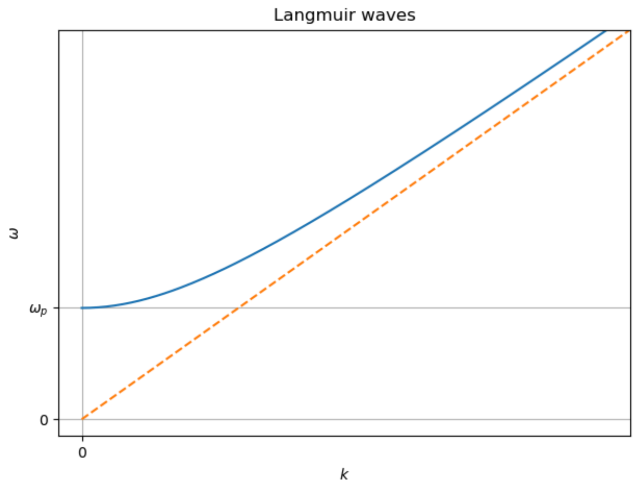
$$\frac{v_{th\,e}^2}{\omega_p^2}=\frac{k_B T_{e0}\,\cancel{m_e}}{\cancel{m_e}\,4\pi n_{e0} e^2}\equiv\lambda_D^2$$

which is the Debye lenght
we also have the group speed

$$v_g=\frac{\partial\omega}{\partial k}=f(k)\neq 0$$

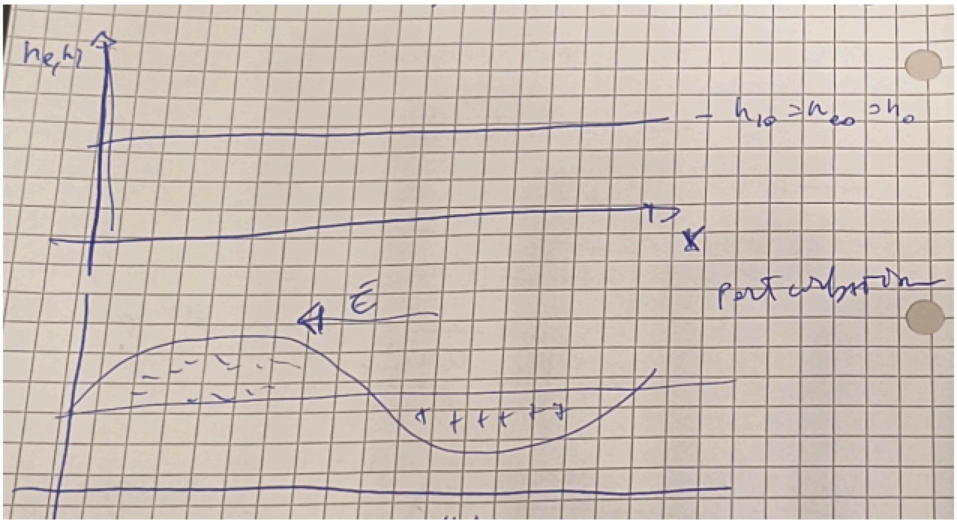
which means that we have dispersive waves.

these are the **Langmuir waves**
pic 01 - L waves



In 1 dimension we have the following oscillatory motion of the system

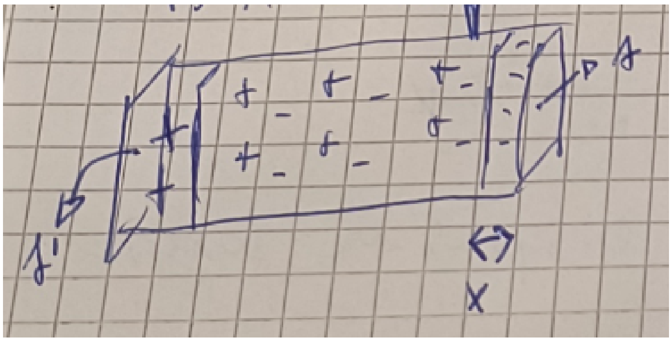
pic 02 - oscill sys



We take a box of ions and electrons

We move the electron cloud "box" to the left

pic 03 - e- cloud



$$E = \frac{\sigma}{\varepsilon_0}, \qquad E = 4\pi\sigma$$

$$n_0 = n_{e\,0} = n_{i\,0}$$

the total charge is

$$Q = |-n_0 A x e|$$

the surface charge is

$$\sigma = \frac{Q}{A} = n_0 x e$$

$$F = -eE = -4\pi n_0 e^2 x$$

$$m\frac{d^2}{dt^2}x = -\left(\frac{4\pi n_0 e}{m}\right)^2 x \implies \frac{d^2x}{dt^2} + \omega_p^2 x = 0$$

Low frequency regime

$$\frac{\partial n_\alpha}{\partial t} + \nabla\cdot\left(n_\alpha \bar{U}_\alpha\right) = 0$$

$$m_\alpha n_\alpha \left[\frac{\partial \bar{U}}{\partial t} + \bar{U}_\alpha \cdot \nabla \bar{U}_\alpha\right] = -\nabla p_\alpha - q_\alpha n_\alpha \nabla\Phi$$

$$-\nabla\Phi = 4\pi \sum_\alpha q_\alpha n_\alpha$$

$$p_\alpha = n_\alpha k_B T_\alpha$$

$$T_{\alpha 0} = const$$

(isothermal EoS)
ions can move
assumption: $m_e \to 0$, so the second equation for electrons becomes

$$0 = -\nabla n_e k_B T_{e0} - q_e n_e \nabla\Phi$$

$$0 = -\frac{1}{n_e}\nabla k_B T_{e0} n_e + e\nabla\Phi =$$

$$= -\nabla k_B T_{e0} \ln n_e + e\nabla\Phi =$$

$$= \nabla\left(e\Phi - k_B T_e \ln n_e\right)$$

$$\implies \quad e\Phi - k_B T_e \ln n_e \equiv \beta$$

with β homogeneous in space
evaluating β at equilibrium

$$\cancel{e\Phi_0} - k_B T_e \ln n_{e0} = \beta \quad \implies \quad \beta = -k_B T_{e0} \ln n_{e0}$$

using again the def of β

$$e\Phi = k_B T_{e0}\left(\ln n_e - \ln n_{e0}\right) \quad \implies \quad \frac{e\Phi}{k_B T_{e0}} = \ln\left(\frac{n_e}{n_{e0}}\right)$$

and we isolate n_e

$$n_e = n_{e0}\exp\left(\frac{e\Phi}{k_B T_{e0}}\right) = n_{e0}\exp\left(\frac{e\Phi_1}{k_B T_{e0}}\right) \approx n_{e0}\left(1 + \frac{e\Phi_1}{k_B T_{e0}}\right)$$

so we finally have a formula for n_e
later we wil solve it also or ions

Linear set of equations for protons (cold, $\nabla p_i \ll q_i n_i \nabla\Phi$)

$$
\begin{cases}
\dfrac{\partial}{\partial t}(n_{i1}) + n_{i0}\nabla\cdot\bar{U}_{i1} = 0 & (A) \\
m_i n_{i0}\dfrac{\partial \bar{U}_{i1}}{\partial t} = -en_{i0}\nabla\Phi_1 & (B) \\
-\nabla^2\Phi_1 = 4\pi e\left[n_i - n_e\right] & (C)
\end{cases}
$$

we have for (C) (using $n_{e0} = n_{i0} = n_0$)

$$
-\nabla^2\Phi_1 = 4\pi e\left[n_i - n_e\right] = 4\pi e\left[\cancel{n_{i0}} + n_{i1} - \cancel{n_{e0}} - n_{e1}\right] = 4\pi e\left[n_{i1} - n_{e1}\right]
$$

and using the previous result we have

$$
-\nabla^2\Phi_1 = 4\pi e\left[n_{i1} - n_{e0}\left(1 + \frac{e\Phi_1}{k_B T_{e0}}\right)\right]
$$

we go to fourier space

$$
\begin{cases}
-i\omega\hat{n}_{i1} + n_0\, i\bar{k}\cdot\hat{\bar{U}}_{i1} = 0 & (A) \\
m_i n_{i0}\,(-i\omega)\hat{\bar{U}}_{i1} = -en_{i0}i\bar{k}\,\hat{\Phi}_1 & (B) \\
k^2\hat{\Phi}_1 = 4\pi e\left[\hat{n}_{i1} - n_{e0}\dfrac{e\hat{\Phi}_1}{k_B T_{e0}}\right] & (C)
\end{cases}
$$

From (A)

$$
\bar{k}\cdot\hat{\bar{U}}_{i1} = \frac{\omega}{n_{i0}}\hat{n}_{i1}
$$

From (B)

$$
\omega m_i n_{i0}\bar{k}\cdot\hat{\bar{U}}_{i1} = en_{i0}k^2\hat{\Phi}_1
$$

and using (A) into this eq

$$
\omega m_i n_{i0}\frac{\omega}{n_{i0}}\hat{n}_{i1} = en_{i0}k^2\hat{\Phi}_1
$$

$$
\implies \hat{n}_{i1} = \frac{ek^2}{m_i\omega^2}n_0\hat{\Phi}_1
$$

So we can rewrite (C) as

$$
k^2\hat{\Phi}_1 = 4\pi e\left[\frac{ek^2}{m_i\omega^2}n_0\hat{\Phi}_1 - n_{e0}\frac{e\hat{\Phi}_1}{k_B T_{e0}}\right]
$$

from which

$$
k^2\hat{\Phi}_1\left[1 + \underbrace{\frac{4\pi e^2 n_0}{k^2 k_B T_{e0}}}_{1/k^2\lambda_D^2} - \underbrace{\frac{4\pi n_0 e^2}{m_i\omega^2}}_{\omega_{pi}^2/\omega^2}\right] = 0 \implies k^2\hat{\Phi}_1\left[1 + \frac{1}{k^2\lambda_D^2} - \frac{\omega_{pi}^2}{\omega^2}\right] = 0
$$

so the dispersion relation is

$$\omega^2 = \frac{\omega_{pi}^2}{1 + \frac{1}{k^2\lambda_D^2}} = \frac{\omega_{pi}^2 k^2 \lambda_D^2}{1 + k^2\lambda_D^2}$$

and

$$\omega = \pm\omega_{pi} k \lambda_D \left(1 + k^2\lambda_D^2\right)^{-1/2}$$

note that

$$\omega_{pi}\lambda_D = \sqrt{\frac{\cancel{4\pi n_0 e^2}}{m_i} \cdot \frac{k_B T_{e0}}{\cancel{4\pi n_0 e^2}}} = \sqrt{\frac{k_B T_{e0}}{m_i}} = c_s$$

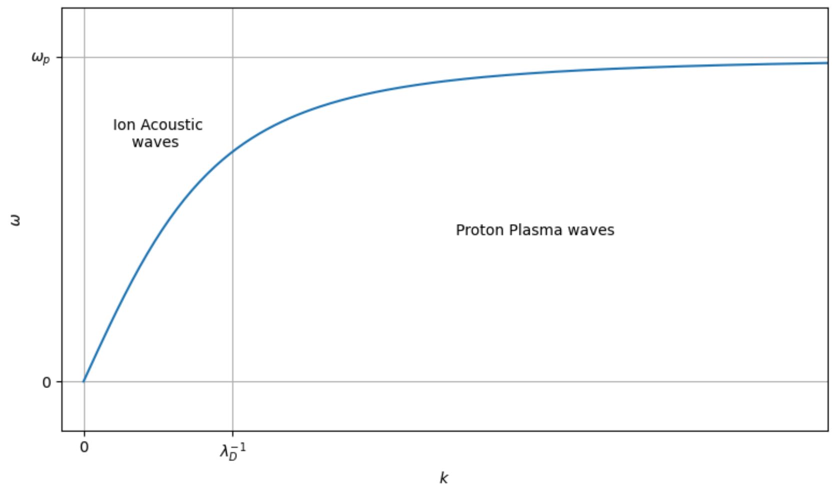
which is the ion sound speed

so

$$\omega = \pm \frac{k c_s}{\sqrt{1 + k^2\lambda_D^2}}$$

defines the **Ion acoustic waves**

pic 04 - Ion acoustic



we can find this kind of waves in the solar wind

Kinetic theory

Hypothesis

- 1D -> $\nabla = \left(\frac{\partial}{\partial x}, 0, 0\right)$
- Electrostatic approximation
- Ion frequency regime, $m_e \to 0$, cold protons

$$\begin{cases} \dfrac{\partial n_i}{\partial t'} + \dfrac{\partial}{\partial x'}(n_i v_i) = 0 & (A) \\ m_i n_i \left[\dfrac{\partial v_i}{\partial t'} + v_i \dfrac{\partial v_i}{\partial x'}\right] = -e n_i \dfrac{\partial \Phi}{\partial x'} & (B) \\ -\dfrac{\partial^2 \Phi}{\partial x'^2} = 4\pi e \left[n_i - n_e\right] & (C) \\ n_e = n_{e0} \exp\left(\dfrac{e\Phi}{k_B T_{e0}}\right) & (D) \end{cases}$$

with

$$n_0 = n_{e0} = n_{i0}, \qquad \Phi_0 = 0, \qquad v_{i0} = 0$$

we use (from ion acoustic case)

- the typical lenght (λ_D)
- the typical speed (c_s)
- the typical frequency (ω_{pi})
  to change variables to adimensional ones
  so

$$v \leftarrow \frac{v'}{c_s}, \qquad x \leftarrow \frac{x'}{\lambda_D}, \qquad \omega \leftarrow \frac{\omega'}{\omega_{pi}}$$

  and

$$t \leftarrow t'\omega_{pi}, \qquad n_i \leftarrow \frac{n_i'}{n_0}, \qquad \phi \leftarrow \frac{e\Phi}{k_B T_{e0}}$$

  From (A)

$$\frac{\partial(n n_0)}{\partial\left(t/\omega_{pi}\right)} + \frac{\partial(n n_0 \cdot v c_s)}{\partial(x \lambda_D)} = 0$$

$$\Longrightarrow \cancel{\omega_{pi} n_0} \frac{\partial n}{\partial t} + \cancel{\frac{n_0 c_s}{\lambda_D}} \frac{\partial(nv)}{\partial x} = 0$$

$$\Longrightarrow \frac{\partial n}{\partial t} + \frac{\partial(nv)}{\partial x} = 0$$

  From (B)

$$m_i n n_0 \left[\frac{\partial(v c_s)}{\partial(t/\omega_{pi})} + v c_s \frac{\partial(v c_s)}{\partial(x\lambda_D)}\right] = -e n n_0 \frac{\partial(\phi \cdot k_B T_{e0}/e)}{\partial(x\lambda_D)}$$

$$\Longrightarrow m_i n \left[\omega_{pi} c_s \cdot \frac{\partial v}{\partial t} + \underbrace{\frac{c_s^2}{\lambda_D}}_{c_s\omega_{pi}} \cdot v \frac{\partial v}{\partial x}\right] = -\frac{\cancel{e} n}{\lambda_D} \frac{k_B T_{e0}}{\cancel{e}} \frac{\partial \phi}{\partial x} = -n \frac{k_B T_{e0}}{\lambda_D} \frac{\partial \phi}{\partial x}$$

$$\Longrightarrow \cancel{n c_s \omega_{pi}} \left[\frac{\partial v}{\partial t} + v \frac{\partial v}{\partial t}\right] = \cancel{n c_s \omega_{pi}} \frac{\partial \phi}{\partial x}$$

  (remembering $c_s^2 = k_B T_{e0}/m_i$)

From (C)

$$
\begin{aligned}
& -\frac{\partial^2}{\partial(x\lambda_D)^2}\left(\frac{k_B T_{e0}}{e}\phi\right) = 4\pi e\left[n_i n_0 - \frac{n_e}{n_0} n_0\right] \\
\implies & -\frac{k_B T_{e0}}{e\lambda_D^2}\frac{\partial^2\phi}{\partial x^2} = 4\pi e^2 n_0\left[n_i - \exp(\phi)\right] \\
\implies & -\cancel{\frac{1}{\lambda_D^2}}\frac{\partial^2\phi}{\partial x^2} = \underbrace{\left(\cancel{\frac{4\pi e^2 n_0}{k_B T_{e0}}}\right)}_{1/\lambda_D^2}\left[n_i - \exp(\phi)\right]
\end{aligned}
$$

Finally

$$
\begin{cases}
\dfrac{\partial n}{\partial t} + \dfrac{\partial(nv)}{\partial x} = 0 & (A) \\
\dfrac{\partial v}{\partial t} + v\dfrac{\partial v}{\partial x} = -\dfrac{\partial\phi}{\partial x} & (B) \\
-\dfrac{\partial^2}{\partial x^2}\phi = n - \exp(\phi) & (C)
\end{cases}
$$

Solitons

They are solutions of constant velocity, fixed shape and non dispersive wave.
pic 05 - solitons

We use $\xi = x - Mt$ with $M = v_0/c_s$ being the Mach speed
for $\xi \to \infty$ we have (with $()' \to \frac{d}{dt}$)

$$
\begin{aligned}
\phi' &= 0, \qquad \phi = 0 \\
v' &= 0, \qquad v = 0 \\
n' &= 0, \qquad n = 1
\end{aligned}
$$

and the equations become

- (A)

$$
\begin{aligned}
& \frac{\partial n}{\partial\xi}\frac{\partial\xi}{\partial t} + \frac{\partial(nv)}{\partial\xi}\frac{\partial\xi}{\partial x} = 0 \\
\implies & -M\frac{\partial n}{\partial\xi} + \frac{\partial(nv)}{\partial\xi} = 0
\end{aligned}
$$

- (B)

$$
-M\frac{\partial v}{\partial\xi} + v\frac{\partial v}{\partial\xi} = -\frac{\partial\phi}{\partial\xi}
$$

- (C)

$$
-\frac{\partial^2\phi}{\partial\xi^2} = n - \exp(\phi)
$$

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